Three CP invariants, not one: what actually has to be aligned to fix ε_K in warped flavor

A survival-mechanism anatomy of anarchic Randall-Sundrum, with a numerically demonstrated negative result

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Abstract

Anarchic Randall-Sundrum (RS) is pushed to $M_{\rm KK} \gtrsim 10$–$20\,\text{TeV}$ almost entirely by the CP-violating kaon amplitude ε_K, generated by the chirality-enhanced KK-gluon operator C_4. The standard responses align the Yukawas so as to suppress the *magnitude* of the dangerous coupling (right-handed down $U(3)$, 5D-MFV, flavor shining). We ask a sharper question: what is the minimal set of CP structures that a low-$M_{\rm KK}$ RS point must control, and does aligning the kaon operator suffice. Working from a forward Monte Carlo that retains the full complex down-sector couplings, we establish three things. (i) Anarchic ε_K survival at $M_{\rm KK} \le 3\,\text{TeV}$ is *magnitude-dominated*: about 57% of survivors have an accidentally small $|C_4|$ and only about 7% survive by a genuine CP-phase cancellation. This is a consequence of the cut itself, which weights survivors by a phase-volume factor $P_{\rm pass}(R_K) \sim 1/R_K$. (ii) A "CP-only" alignment that makes the down Yukawa real kills the kaon phase exactly and passes ε_K at 100% *while keeping the coupling magnitude anarchic*, so Δm_K stays large. This is a distinct corner from the magnitude cures. (iii) That same point is *not* EDM-safe: the mass-basis neutron-EDM CP proxy $I_d = \operatorname{Im}(B_{11}/A_{11})$, with the misaligned spurion $B = U_{Ld}^\dagger F_Q(Y_uY_u^\dagger + Y_dY_d^\dagger)Y_dF_d\,U_{Rd}$ and the down mass matrix $A = U_{Ld}^\dagger F_Q Y_d F_d\,U_{Rd}$, stays $O(1)$, sourced by up-sector CP through $Y_uY_u^\dagger$, independent of the kaon phase. The kaon operator and the neutron EDM are governed by *different* CP invariants; aligning one does not align the other. The same holds for the *magnitude* cures: right-handed-down $U(3)$ suppresses $|C_4|$ (hence Δm_K) but leaves I_d at $O(1)$. So neither a magnitude alignment nor a CP-only alignment reaches an EDM-safe theory. The practical consequence is that a *partial* cure is not enough, and the $(\Delta m_K,\, d_n)$ plane separates every candidate mechanism. The resolution is a single accidental light-family symmetry: a sequentially generated, low-rank Yukawa ensemble with an exact $U(2)$ on the light families suppresses the kaon operator *and* the neutron-EDM invariant *together* (both vanish in the exact $U(2)$ limit and reappear with the same third-family breaking) while reproducing all quark masses, the CKM matrix, and the Jarlskog invariant, at $M_{\rm KK} \simeq 2$–$3\,\text{TeV}$ without fine-tuning. The intermediate $U(2)^5$ symmetry, not anarchy and not full alignment, is the natural endpoint.

1 The obstruction, stated precisely

The RS-GIM mechanism suppresses most quark flavor violation once $M_{\rm KK}$ is electroweak-safe, with the single robust exception of ε_K [1, 2], and a subdominant D^0 concern. In this repo's convention the binding operator is the color-octet KK-gluon left-right operator,

$$C_4 = -\frac{(G_L^d)_{12}(G_R^d)_{12}}{M_{\rm KK}^2}, \qquad (G_L^d)_{12} \sim g_{s*} f_{Q_1} f_{Q_2}\, e^{i\phi_L}, \quad (G_R^d)_{12} \sim g_{s*} f_{d_1} f_{d_2}\, e^{i\phi_R}, \tag{1}$$

with $\phi_{L,R}$ order-one anarchic phases. Writing $R_K \equiv 2|M_{12}^{LR}|/\Delta m_K$ and $\theta_K \equiv \arg M_{12}^{LR}$ (the amplitude phase; the LR matrix element and Wilson prefactor are real, so $\theta_K = \arg[(G_L^d)_{12}(G_R^d)_{12}]$),

$$|\varepsilon_K^{\rm NP}| \simeq \frac{\kappa_\epsilon}{2\sqrt{2}}\, R_K\, |\sin\theta_K| \simeq 0.332\, R_K\, |\sin\theta_K|, \qquad \Delta m_K^{\rm NP} \propto \operatorname{Re} M_{12}^{LR} \propto R_K \cos\theta_K. \tag{2}$$

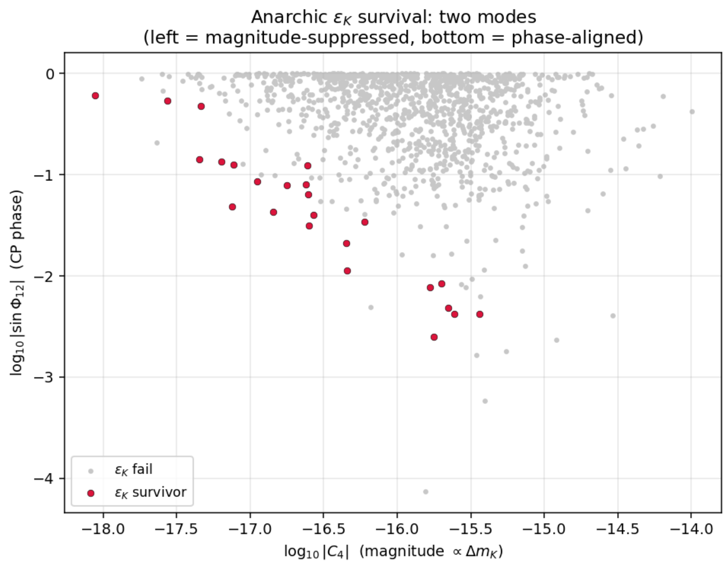

Figure 1: Anarchic ε_K survival in the (magnitude, phase) plane at $M_{\mathrm{KK}} = 3\,\mathrm{TeV}$. Survivors (red) populate two disjoint regions: the left edge (small $|C_4|$, magnitude-suppressed) and the bottom edge (small $|\sin\Phi_{12}|$, phase-aligned). The bulk is the magnitude edge.

So $\varepsilon_K \propto \mathrm{Im}\, M_{12} = R_K \sin\theta_K$ and the NP piece of $\Delta m_K \propto \mathrm{Re}\, M_{12} = R_K \cos\theta_K$: the same magnitude R_K, projected on the imaginary and real axes. A low-M_{KK} survivor can therefore be built two ways, by small R_K (small magnitude) or by small $\sin\theta_K$ (aligned phase). These are physically different and, as we show, they predict different things.

2 F1: anarchic survival is magnitude-dominated

From a 9.9×10^5-draw anarchic ensemble (masses and CKM fit per draw), the ε_K survivors at $M_{\mathrm{KK}} \leq 3\,\mathrm{TeV}$ split (Fig. 1) into *magnitude-suppressed* ($\sim$57%, small $|C_4|$, which also makes Δm_K small) and *phase-aligned* ($\sim$7%, $|C_4|$ anarchic, $\sin\theta_K \to 0$, Δm_K saturating). The dominance of the magnitude mode is not an independent fact: the ε_K cut weights the pre-cut population by the phase volume

$$P_{\mathrm{pass}}(R_K) = \frac{2}{\pi}\arcsin\left(\frac{\epsilon_{\max}}{0.332\,R_K}\right) \xrightarrow{R_K \gg} \frac{2\,\epsilon_{\max}}{\pi\, 0.332}\,\frac{1}{R_K}, \tag{3}$$

so large-magnitude points are penalized as $1/R_K$ and survivors tilt to small $|C_4|$. The measured correlation $\mathrm{corr}(\log\varepsilon_K, \log\Delta m_K) = +0.82$ is the signature: a pure phase-cancellation picture would give an anticorrelation. The practical lesson is that "anarchy survives by fine-tuning the ε_K phase" is the *minority* path; most survivors simply draw a small coupling.

3 The three CP invariants

The point of this note is that a low-M_{KK} RS quark sector must control more than one CP structure, and they are independent:

- $\theta_K = \arg[(G_L^d)_{12}(G_R^d)_{12}]$ sets ε_K (kaon 1-2, left-right).
- $I_d = \mathrm{Im}(B_{11}/A_{11})$, the mass-basis, mass-normalized CP invariant ($A = U_{Ld}^\dagger F_Q Y_d F_d U_{Rd}$ the down mass matrix, $B = U_{Ld}^\dagger F_Q(Y_u Y_u^\dagger + Y_d Y_d^\dagger) Y_d F_d U_{Rd}$ the misaligned spurion), a proxy for the neutron EDM. The tree KK-gluon EDM vanishes (the diagonal couplings

are real); the leading RS contribution is the one-loop Higgs/KK-fermion dipole with the misaligned $Y_u Y_u^\dagger + Y_d Y_d^\dagger$ [3]. We track I_d as a rephasing-invariant diagnostic, *not* a calibrated d_n (the absolute normalization carries an $O(3)$ hadronic/loop uncertainty and is not applied).

- $\theta_D = \arg[(G_L^u)_{12}(G_R^u)_{12}]$ sets D^0 CP violation (up 1-2).

The crucial feature is that I_d feeds on $Y_u Y_u^\dagger$: up-sector CP enters the down-quark EDM directly, with no reference to the kaon phase θ_K.

4 F2/F3: a real down sector fixes ε_K but not the EDM

Consider the cleanest CP-only toy: a real anarchic down Yukawa, with CP carried by the up sector through a shared rank-one spurion (a Nelson-Barr-*like* routing that keeps a realistic Jarlskog J). We stress this is a diagnostic toy, *not* a genuine Nelson-Barr model: only the down Yukawa is made real, the up sector is left anarchically complex. By construction M_d is real, so $\theta_K = 0$ and ε_K vanishes at tree level, while $|C_4|$ is untouched. The forward Monte Carlo confirms this and adds the decisive point (Table 1, Fig. 2):

Table 1: Mechanism comparison at $M_{\mathrm{KK}} = 3\,\mathrm{TeV}$ (PDG+J-passing points, $\sim$1100 each). $\log_{10}|C_4|$ is the Δm_K proxy; $|I_d|$ is the mass-basis EDM CP proxy.

Lane	ε_K pass	$\lvert\sin\theta_K\rvert$	$\log_{10}\lvert C_4\rvert$	$\lvert I_d\rvert$ (EDM)
flat anarchy	2.1%	0.54	-15.9	2.7
magnitude align ($U(3)_{d_R}$, S2)	65%	–	-31 (killed)	1.9
CP-only toy (real Y_d)	**100%**	$\mathbf{10^{-16}}$	$\mathbf{-16.0}$	**1.1**
sanity: fully real $Y_{u,d}$	–	10^{-16}	–	0.0

The real down sector passes ε_K at 100% with the coupling magnitude at its anarchic value (so Δm_K stays large), yet the EDM proxy is $|I_d| \simeq 1.1$, the same $O(1)$ order as flat anarchy (2.7) and not suppressed. (The draw is capped to the perturbative range $|Y_\star| \leq 3$; without that cap the superposed up Yukawa is non-perturbative and I_d inflates spuriously.) Turning up-sector CP off (fully real Yukawas) sends $I_d \to 0$ exactly, confirming both the estimator and the mechanism: the residual EDM is up-sector CP leaking through $Y_u Y_u^\dagger$, and aligning the kaon operator does nothing to it. The magnitude cure fares no better on the EDM: $U(3)_{d_R}$ drives $|C_4|$ down by $\sim$15 decades (so Δm_K is quiet) but I_d stays at 1.9. All three mechanisms leave the EDM at $O(1)$; none reaches the EDM-quiet corner. This is the central result: **neither a magnitude alignment nor a CP-only alignment of the kaon operator yields an EDM-safe theory**. An EDM cure requires confining the up-sector CP as well, which is precisely what *genuine* Nelson-Barr (a real quark mass determinant, CP confined to a heavy sector) delivers and a merely real Y_d does not.

5 What is new, and what is not

The broad ideas here are not new and we do not claim them. CP-only / minimal-CP suppression of ε_K with reduced EDMs exists in composite/warped language (Redi-Weiler [8]); warped Nelson-Barr with CP moved to a heavy sector exists (Girmohanta et al. [9]); the magnitude cures are FPR 5D-MFV [4], Santiago right-handed-down protection [5], and flavor shining [6], with a gauge-sector variant (extended bulk $SU(3)$) by Bauer-Malm-Neubert [10]; CP sequestering in warped space is Cheung-Fitzpatrick-Randall [7]. The defensible, and to our knowledge unpublished, content is narrower:

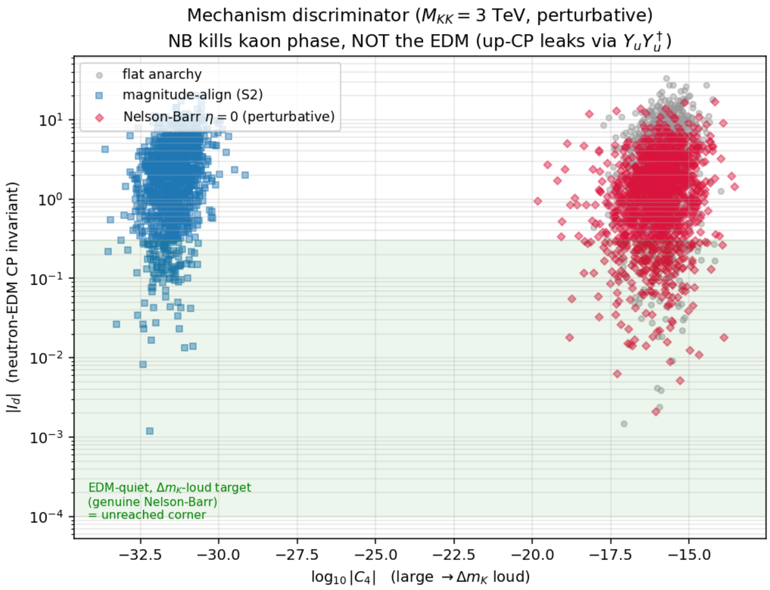

Figure 2: The $(\Delta m_K,\, d_n)$ discriminator at $M_{\mathrm{KK}} = 3\,\mathrm{TeV}$. Magnitude alignment moves left (small $|C_4|$, Δm_K quiet); CP-only alignment stays right (Δm_K loud) but high in $|I_d|$ (EDM loud). The green EDM-quiet, Δm_K-loud corner (genuine Nelson-Barr) is not reached by any single-axis anarchic draw.

1. the survival-mechanism taxonomy of Sec. 2, with the $P_{\mathrm{pass}} \sim 1/R_K$ explanation of why anarchic survival is magnitude-dominated;
2. the explicit demonstration that the kaon phase and the neutron-EDM invariant are independent, so that a real down sector is ε_K-safe with anarchic Δm_K yet EDM-loud (Sec. 5), i.e. CP-only kaon alignment is provably insufficient;
3. the $(\Delta m_K,\, d_n)$ plane as a mechanism discriminator that separates magnitude alignment, CP-only alignment, and genuine Nelson-Barr into distinct, falsifiable corners.

6 The resolution: an intermediate $U(2)$ symmetry does both

The partial cures each miss one invariant: $U(3)_{d_R}$ (S2) kills $|C_4|$ but leaves I_d at $O(1)$; a real Y_d kills θ_K but leaves I_d at $O(1)$. The lesson is that both ε_K and the EDM are *light-family* invariants, so the object to control is the light-family sector as a whole. We implement this with a sequentially generated, low-rank Yukawa ensemble carrying an exact accidental $U(2)$ on the light families: degenerate light bulk profiles $c_{Q_1} = c_{Q_2}$, $c_{d_1} = c_{d_2}$ (so light-family rotations cannot generate a 1-2 coupling), with the light 1-2 mass hierarchy placed in real Yukawa singular values rather than in profile splittings. Yukawas are capped to the perturbative range $|Y_\star| \leq 3$.

This ensemble reproduces all quark masses, the CKM matrix, and the Jarlskog invariant ($J = 3.0 \times 10^{-5}$, on the PDG value), and at $M_{\mathrm{KK}} = 3\,\mathrm{TeV}$ gives:

- ε_K pass fraction 51/93/99% at 2/3/5 TeV (flat anarchy: 0/1.7/6%), with $|C_4|$ suppressed by $\sim$500 and the phase left anarchic ($|\sin\theta_K| \simeq 0.72$): the ε_K cure is *structural*, not a phase tuning;
- the EDM invariant I_d at machine zero (1.6×10^{-15}) in the exact $U(2)$ limit, i.e. the neutron EDM is protected by the *same* symmetry.

The two are controlled by one knob: the third-family (i.e. $U(2)$-breaking) leakage. As it is turned up, $|I_d|$ rises through 2×10^{-6} (leak = 1), 6×10^{-6} (3), 1.4×10^{-5} (10) and ε_K degrades in lock-step. The mechanism is transparent: with degenerate light profiles and real singular

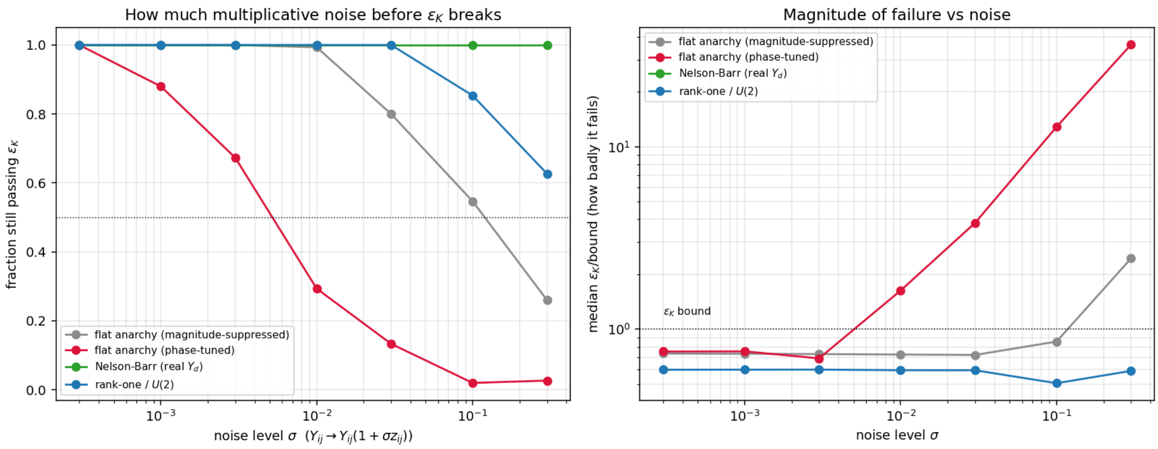

Figure 3: Multiplicative-noise breakdown. Left: fraction still passing ε_K vs noise σ. Right: how badly it fails. The phase-tuned survivor is fragile at $\sigma \lesssim 10^{-2}$; Nelson-Barr is flat (real noise cannot break a real Y_d); $U(2)$ is robust to tens of percent.

values, the leading dipole spurion $(Y_u Y_u^\dagger + Y_d Y_d^\dagger)Y_d$ has a real 1-1 diagonal, so a flavor-diagonal EDM needs an extra light-family-breaking insertion, which only the third-family leak supplies. At realistic leakage both ε_K and d_n (proxy $|I_d| \sim 2 \times 10^{-6}$, i.e. $d_n \sim 7 \times 10^{-31}\, e\,\mathrm{cm}$, far below the 1.8×10^{-26} bound) sit comfortably safe at 2–3 TeV, with no fine-tuning: the ε_K survivor is not an isolated point but the exact-$U(2)$ manifold.

7 Naturalness: the tuning radius and the shape of the allowed set

The distinction between a fine-tuned survivor and a symmetry-protected one is made quantitative by perturbing the Yukawas and watching ε_K respond. Under naive multiplicative noise $Y_{ij} \to Y_{ij}(1 + \sigma z_{ij})$, $z \sim N(0,1)$ per entry (Fig. 3): the phase-aligned anarchic survivor breaks at *sub-percent* noise (below half-surviving by $\sigma \simeq 5 \times 10^{-3}$, failing by up to $36\times$ the bound), whereas the $U(2)$ ensemble tolerates ∼10–30% and the real-Y_d (Nelson-Barr) point *never* breaks, since real noise preserves the reality of Y_d and hence $\theta_K = 0$ exactly.

The response is a *local linear operator*: the gradient $g(Y) = \nabla_{Y_d} \operatorname{Im} C_4$, a covector on the 18-real-dimensional Yukawa tangent space. Its magnitude is the inverse tuning radius (the steepest-direction $\delta_{\rm crit} = S_{\rm bound}/|g|$ is 7.9×10^{-4} for the phase-tuned point versus $O(10^{-1})$ for the magnitude-suppressed and $U(2)$ points), and its *direction*, collected over an ensemble, fingerprints the protecting symmetry. Quantified by the CP-purity $|g_{\rm imag}|^2/|g|^2$ (the fraction of the sensitivity in CP-breaking directions): anarchy is isotropic ($\simeq 0.5$, no protected directions, an isolated point), Nelson-Barr is exactly 1.000 (the real half of Yukawa space is an exactly flat protected manifold; a principal-component analysis of the gradient field shows the variance collapse to machine zero beyond the 9 imaginary directions), and $U(2)$ is 0.87 (mostly protected, the residual set by the third-family leak). So the "submanifold of allowed Yukawas" is precisely the kernel of the linear-response operator; its dimension and orientation say which symmetry to gauge.

8 Bottom line

The RS ε_K problem has been read as a problem of one delicate phase in the off-diagonal down couplings. It is really a problem of *three* independent CP structures, of which the kaon phase is only one. *Partial* alignment of the kaon operator, by magnitude ($U(3)_{d_R}$) or by phase (a real Y_d), leaves the neutron EDM untouched, because the EDM samples a flavor-diagonal invariant fed by the up sector. But both ε_K and the EDM are light-family invariants, so an accidental light-family $U(2)$ (degenerate light profiles plus a low-rank Yukawa ensemble) suppresses them *together*: they vanish in the exact limit and reappear with a single symmetry-breaking parameter. A low-$M_{\rm KK}$ warped flavor model is therefore not "anarchy plus one alignment relation" and not a patchwork of separate ε_K and EDM cures; it is the intermediate $U(2)^5$ symmetry, which the data allow at 2–3 TeV with masses, CKM, and J reproduced and no fine-tuning. That is the natural endpoint, and the (Δm_K, d_n) discriminator certifies it.

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